

MongoDB Geospatial Queries

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Overview

- ▶ MongoDB supports geospatial queries on:
 - ▶ GeoJSON objects (spherical geometry)
 - ▶ Legacy coordinate pairs (planar geometry)
- ▶ Geospatial indexes enhance query performance
- ▶ Applications include location-based services, mapping, and spatial analysis

GeoJSON Format

- ▶ Represents geospatial data on an earth-like sphere.
- ▶ MongoDB uses GeoJSON objects to represent geospatial data.
- ▶ Common GeoJSON types:
 - ▶ Point
 - ▶ LineString
 - ▶ Polygon
- ▶ Coordinates are in [longitude, latitude] order.

Example of a GeoJSON Point:

```
{  
  "type": "Point",  
  "coordinates": [-73.856077, 40.848447]  
}
```

Legacy Coordinate Pairs

- ▶ Represents geospatial data on a Euclidean plane
- ▶ Preferred format:

```
"location": [-73.856077, 40.848447]
```

- ▶ Also supports embedded documents:

```
"location": { "x": -73.856077, "y": 40.848447 }
```

- ▶ Use 2d indexes for planar geometry

Geospatial Indexes

▶ 2dsphere Index:

- ▶ Supports spherical geometry
- ▶ Required for GeoJSON objects
- ▶ Created with:

```
db.collection.createIndex({ location: "2dsphere" })
```

▶ 2d Index:

- ▶ Supports planar geometry
- ▶ Suitable for legacy coordinate pairs
- ▶ Created with:

```
db.collection.createIndex({ location: "2d" })
```

Geospatial Query Operators

- ▶ **\$near:**
 - ▶ Finds documents close to a point
 - ▶ Requires a geospatial index
- ▶ **\$geoWithin:**
 - ▶ Finds documents within a specified geometry
- ▶ **\$geoIntersects:**
 - ▶ Finds documents that intersect with a geometry
- ▶ **\$nearSphere:**
 - ▶ Similar to **\$near** but calculates distances on a sphere

Example: \$near Query

```
1  db.places.find({
2    location: {
3      $near: {
4        $geometry: {
5          type: "Point",
6          coordinates: [-73.856077, 40.848447]
7        },
8        $maxDistance: 1000
9      }
10   }
11 })
```

- Finds places within 1 km of the specified point

Example: \$geoWithin Query

```
1  db.places.find({
2      location: {
3          $geoWithin: {
4              $geometry: {
5                  type: "Polygon",
6                  coordinates: [[
7                      [-73.958, 40.800],
8                      [-73.949, 40.796],
9                      [-73.973, 40.764],
10                     [-73.981, 40.768],
11                     [-73.958, 40.800]
12                 ]]
13             }
14         }
15     })
```

- Finds places within the specified polygon

Geospatial Indexing

- ▶ Introduction to MongoDB's geospatial indexing capabilities.
- ▶ Demonstration of geospatial queries using:
 - ▶ `$geoWithin`
 - ▶ `$geoIntersects`
 - ▶ `$nearSphere`
- ▶ Use case: Finding restaurants in New York City.

Geospatial Indexes

- ▶ MongoDB supports geospatial indexes to efficiently execute spatial queries.
- ▶ Types of geospatial indexes:
 - ▶ **2d** index: For planar geometry.
 - ▶ **2dsphere** index: For spherical geometry (e.g., Earth).
- ▶ This tutorial uses **2dsphere** indexes.

Importing Data

- ▶ Download datasets:
 - ▶ `restaurants.json`
 - ▶ `neighborhoods.json`
- ▶ Import into MongoDB:

```
mongoimport <path to restaurants.json> -c=restaurants  
mongoimport <path to neighborhoods.json> -c=neighborhoods
```

Creating Geospatial Indexes

- Create 2dsphere indexes on the collections:

```
db.restaurants.createIndex({ location: "2dsphere" })  
db.neighborhoods.createIndex({ geometry: "2dsphere" })
```

Sample Restaurant Document

```
{
  "location": {
    "type": "Point",
    "coordinates": [-73.856077, 40.848447]
  },
  "name": "Morris Park Bake Shop"
}
```

Sample Neighborhood Document

```
{
  "geometry": {
    "type": "Polygon",
    "coordinates": [[
      [-73.99, 40.75],
      [-73.98, 40.76],
      [-73.99, 40.75]
    ]]
  },
  "name": "Hell's Kitchen"
}
```

Finding Current Neighborhood

- Determine the user's neighborhood using `$geoIntersects`:

```
db.neighborhoods.findOne({
  geometry: {
    $geoIntersects: {
      $geometry: {
        type: "Point",
        coordinates: [-73.93414657, 40.82302903]
      }
    }
  }
})
```

Counting Restaurants in Neighborhood

- Find all restaurants within the user's neighborhood:

```
var neighborhood = db.neighborhoods.findOne({
  geometry: {
    $geoIntersects: {
      $geometry: {
        type: "Point",
        coordinates: [-73.93414657, 40.82302903]
      }
    }
  }
})
db.restaurants.find({
  location: {
    $geoWithin: {
      $geometry: neighborhood.geometry
    }
  }
}).count()
```

Finding Nearby Restaurants (Unsorted)

- Use `$geoWithin` with `$centerSphere` to find restaurants within 5 miles:

```
db.restaurants.find({  
  location: {  
    $geoWithin: {  
      $centerSphere: [[-73.93414657, 40.82302903], 5 / 3963.2]  
    }  
  }  
})
```

Note: `$centerSphere`'s second argument accepts the radius in radians, so you must divide it by the radius of the earth in miles.

Finding Nearby Restaurants (Sorted)

- Use `$nearSphere` with `$maxDistance` to find and sort restaurants by proximity:

```
var METERS_PER_MILE = 1609.34
db.restaurants.find({
  location: {
    $nearSphere: {
      $geometry: {
        type: "Point",
        coordinates: [-73.93414657, 40.82302903]
      },
      $maxDistance: 5 * METERS_PER_MILE
    }
  }
})
```

Summary

- ▶ MongoDB's geospatial features enable efficient spatial queries.
- ▶ Utilizing `2dsphere` indexes and GeoJSON formats allows for complex geospatial operations.
- ▶ Operators like `$geoWithin`, `$geoIntersects`, and `$nearSphere` provide powerful querying capabilities.
- ▶ MongoDB provides robust support for geospatial data
- ▶ Choose appropriate data formats and indexes based on your application's needs
- ▶ Utilize geospatial query operators to perform spatial searches

Resources

- ▶ Official Documentation: [MongoDB Geospatial Queries](#)
- ▶ MongoDB Geospatial Tutorial:
<https://www.mongodb.com/docs/manual/tutorial/geospatial-tutorial/>
- ▶ GeoJSON Specification: geojson.org